Proposal Title Placeholder

FI: M. Bronco — **Advisor:** Prof. KS
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Project Summary

Overview: Provide a concise overview of the research problem, state of knowledge, central gap, hypothesis or design claim, objectives, and general methodology.

Intellectual Merit & Contributions: Detail the primary intellectual contributions of the proposed dissertation research, distinguishing novel theoretical or engineering findings from standard deliverables.

Broader Significance: Explain why resolving this problem matters within ECE and its broader societal, technological, or economic impacts.

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December 2026

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1 Motivation and Significance

Define the consequential scientific or technological problem and the context in which it occurs. Explain why resolving the problem matters and establish the boundaries of the proposed dissertation.

Based on results by Bronco *et al* [1],...

2 State of Knowledge and Research Gap

Synthesize the most relevant literature by claims, mechanisms, approaches, or limitations. Establish what is known, identify where evidence is incomplete or conflicting, and define the specific unresolved gap addressed by the dissertation.

3 Central Question, Hypothesis/Design Claim, and Objectives

State the organizing research question or engineering challenge, central hypothesis or design claim, and two to four coherent research objectives. Explain how completing the objectives will address the central gap.

Objective 1: Develop...

Objective 2: Validate...

Objective 3: Implement...

4 Preliminary Work and Readiness

Present preliminary evidence that establishes feasibility, validates a method, motivates a design decision, or refines the proposed direction. Identify limitations and distinguish the student's contributions from those of collaborators.

5 Research Plan

For each objective, present the rationale, research question or claim, methods, variables and controls or baselines, analysis, uncertainty, success criteria, expected outcomes, risks, alternatives, and objective-level deliverables.

5.1 Objective 1 Methodology

Detail research activities and experimental design...

5.2 Objective 2 Methodology

Detail frameworks and theoretical modeling...

6 Integration and Contributions

Explain how the objectives collectively address the central gap. Distinguish research products—such as devices, algorithms, datasets, models, and publications—from the dissertation's enduring intellectual contribution.

7 Timeline, Risks, and Completion Plan

Present a two- to three-year timeline containing research milestones, dependencies, decision points, analysis, writing, and dissemination. Identify consequential risks, observable triggers, mitigation strategies, and alternative paths.

Table 1: Risk Register, Triggers, Mitigation Strategies, and Alternative Paths

Risk	Observable Trigger	Mitigation Strategy	Alternative Path
High edge latency	Processing time ¿ 15ms	Optimize matrix kernels in FPGA	Fallback to lower-order estimation
Data dropouts	Packet loss ¿ 5%	Implement local Kalman prediction	Use historical state extrapolation

References

[1] S. Bronco, S. I. Bronco, and M. M, “Development and Validation of Open Source Software for Proposal Writing,” *IEEE*, May 2026.